

```

import argparse
import ipaddress
import socket

PORTS = [21, 22, 80, 139, 443, 445, 1433, 3306, 3389, 8080]

parser = argparse.ArgumentParser(description="TCP Port Scanner")

parser.add_argument("subnet")

args = parser.parse_args()

network = ipaddress.ip_network(args.subnet, strict=False)

socket.setdefaulttimeout(0.2)

print(f"Scanning network {network}\n")

for ip in network.hosts():
    ip = str(ip)

    open_ports = []

    for port in PORTS:
        sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

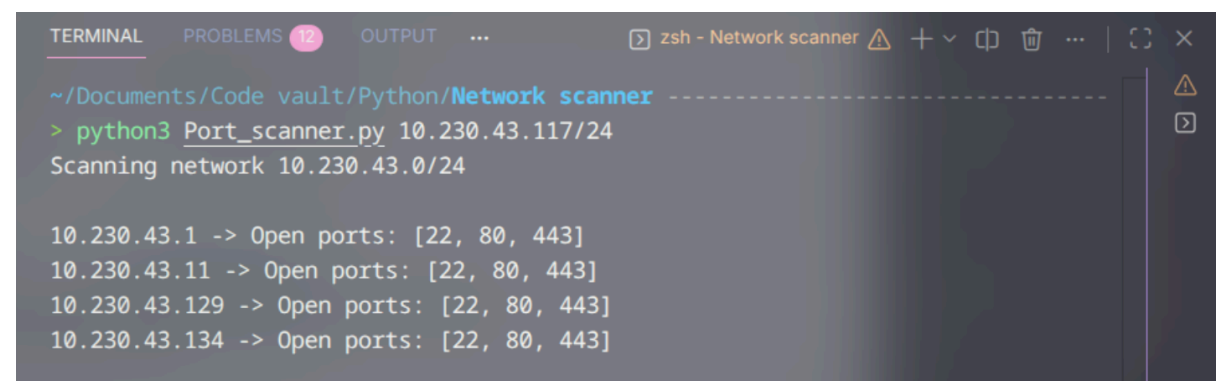
        result = sock.connect_ex((ip, port))

        sock.close()

        if result == 0:
            open_ports.append(port)

    if open_ports:
        print(f"{ip} -> Open ports: {open_ports}")

```



The screenshot shows a terminal window titled "zsh - Network scanner". The user is in the directory `~/Documents/Code vault/Python/Network scanner`. They have run the command `python3 Port_scanner.py 10.230.43.117/24`. The script outputs "Scanning network 10.230.43.0/24" and then lists the open ports for four IP addresses in the range: 10.230.43.1, 10.230.43.11, 10.230.43.129, and 10.230.43.134. All of them show open ports at 22, 80, and 443.

```

TERMINAL  PROBLEMS 12  OUTPUT  ...  zsh - Network scanner
~/Documents/Code vault/Python/Network scanner
> python3 Port_scanner.py 10.230.43.117/24
Scanning network 10.230.43.0/24

10.230.43.1 -> Open ports: [22, 80, 443]
10.230.43.11 -> Open ports: [22, 80, 443]
10.230.43.129 -> Open ports: [22, 80, 443]
10.230.43.134 -> Open ports: [22, 80, 443]

```